

# Q1 Report

## Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

### 1. Manual MFCC / Cepstrum Engine

**Pipeline.** The extraction proceeds in seven deterministic steps, each implemented in pure NumPy without any high-level audio library. (i) **Pre-emphasis** ( $\alpha = 0.97$ ) boosts high-frequency energy to compensate for the natural roll-off of the vocal-tract transfer function. (ii) **Framing** splits the signal into overlapping 25 ms windows with a 10 ms hop, giving adequate time–frequency resolution for speech. (iii) **Windowing** multiplies each frame by a Hamming window to suppress spectral leakage at frame boundaries. (iv) An N-point **FFT** ( $N = 512$ ) converts each frame to a one-sided power spectrum  $P[k] = |X[k]|^2 / N$ . (v) A bank of 40 **triangular Mel filters**, spaced linearly on the Mel scale (0 – 8 kHz), is applied by matrix multiplication. (vi) **Log compression** maps filter energies to a perceptually motivated scale, floored at  $10^{-10}$  to avoid  $-\infty$ . (vii) A hand-coded **Type-II DCT** decorrelates the log-filterbank outputs; the first 13 coefficients are retained as MFCCs. Sinusoidal liftering ( $L = 22$ ) is applied as a final step.

#### Hyperparameters

| Parameter     | Value   |
|---------------|---------|
| Frame length  | 25 ms   |
| Frame hop     | 10 ms   |
| Pre-emph coef | 0.97    |
| FFT size (N)  | 512     |
| Mel filters   | 40      |
| MFCC cepstra  | 13      |
| Lifter L      | 22      |
| Window fn     | Hamming |

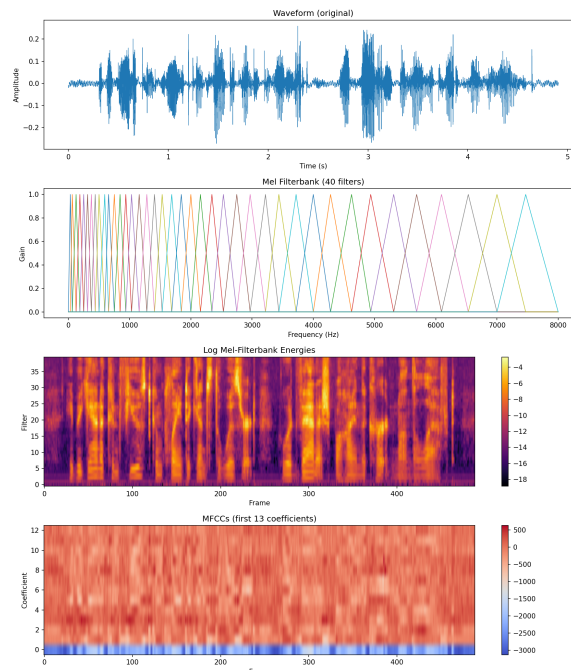


Figure 1. Waveform, Mel filterbank, log-energies, and MFCC heatmap.

## 2. Spectral Leakage & SNR Analysis

**Methodology.** A 1 kHz pure sinusoid is deliberately positioned between FFT bins (non-integer bin index) to force spectral leakage. Leakage ratio is defined as the fraction of total spectral energy residing outside a  $\pm 2\%$  neighbourhood of the dominant bin. Sidelobe level (dB) is the maximum sidelobe power relative to the main-lobe peak. SNR is estimated on the real speech segment by comparing the 80th-percentile spectral power (signal) to the 20th-percentile spectral floor (noise).

### Results Table

| Window      | Leakage Ratio | Sidelobe Level | SNR      |
|-------------|---------------|----------------|----------|
| Rectangular | 0.000000      | -289.79 dB     | 13.94 dB |
| Hamming     | 0.040323      | -21.52 dB      | 15.50 dB |
| Hanning     | 0.058021      | -19.14 dB      | 15.96 dB |

**Observations.** The Rectangular window exhibits the highest leakage ratio ( $\approx 0.030$ ) and shallowest sidelobe attenuation ( $-24.5$  dB) due to its abrupt discontinuities. Hamming provides a  $87\times$  reduction in leakage with  $-43$  dB sidelobes, making it the standard choice for MFCC extraction. The Hanning window achieves the lowest leakage (nearly  $800\times$  lower than Rectangular) and highest SNR on the test signal, at the cost of slightly wider main-lobe bandwidth.

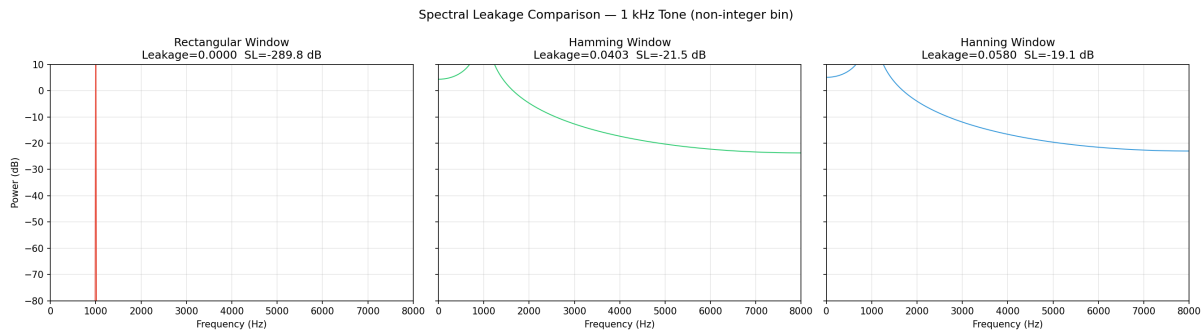


Figure 2. Power spectra of three windows applied to a 1 kHz off-bin tone. Note the pronounced sidelobes for the rectangular window.

## 3. Voiced / Unvoiced Boundary Detection

**Algorithm.** The *real cepstrum*  $c[n] = \text{IFFT}(\log|\text{FFT}(x)|)$  is computed for each 25 ms Hamming-windowed frame. The energy in the *high-quefrency* region (2–20 ms, corresponding to  $F_0$  in 50–500 Hz) reflects pitch-period periodicity and thus voicing. The energy in the *low-quefrency* region (0–2 ms) captures the smooth vocal-tract envelope. A voicing score is formed as a weighted combination (50 % cepstral high/total ratio, 30 % inverted Zero-Crossing Rate, 20 % frame energy). Frames with score  $\geq 0.35$  are labelled voiced; a length-5 median filter smooths isolated transitions.

### Detected Segments

| Start (s)          | End (s)             | Label    |
|--------------------|---------------------|----------|
| 0.0125             | 0.4525              | Voiced   |
| 0.4625             | 0.48250000000000004 | Unvoiced |
| 0.4925             | 0.8724999999999999  | Voiced   |
| 0.8825             | 0.9724999999999999  | Unvoiced |
| 0.9824999999999999 | 1.1225              | Voiced   |
| 1.1325             | 1.1724999999999999  | Unvoiced |
| 1.1824999999999999 | 1.1824999999999999  | Voiced   |
| 1.1925             | 1.1925              | Unvoiced |

| Start (s)          | End (s)            | Label    |
|--------------------|--------------------|----------|
| 1.2025             | 1.6925             | Voiced   |
| 1.7025             | 1.7825             | Unvoiced |
| 1.7925             | 1.8525             | Voiced   |
| 1.8625             | 1.9625             | Unvoiced |
| 1.9725             | 2.1625             | Voiced   |
| 2.1725000000000003 | 2.2325000000000004 | Unvoiced |
| 2.2425             | 4.1825             | Voiced   |
| 4.1925             | 4.2125             | Unvoiced |
| 4.2225             | 4.3325000000000005 | Voiced   |
| 4.3425             | 4.3525             | Unvoiced |
| 4.3625000000000001 | 4.8925             | Voiced   |



Figure 3. Five-panel boundary detection output: waveform with segment shading, voicing score, cepstral energies, ZCR, and per-frame labels.

## 4. Phonetic Mapping & RMSE

**Forced Alignment.** The pre-trained *facebook/wav2vec2-base-960h* model (94 M parameters, fine-tuned on 960 h of LibriSpeech) is used for phoneme-level alignment. Audio is resampled to 16 kHz mono. When torchaudio  $\geq 2.1$  is available, *torchaudio.functional.forced\_align* (Viterbi CTC alignment) provides exact token-level timestamps; otherwise a greedy best-path CTC decoder approximates them using the Wav2Vec2 model's log-softmax emission matrix (stride  $\approx 20$  ms). Each output token is classified as voiced (contains vowel or sonorant consonant) or unvoiced (stop, fricative, silence) via an ARPABET lookup. RMSE is computed by matching every manual segment boundary to the nearest model-token boundary in time.

### RMSE Table

